

Waveform Presentation Guide — SIH26181

Quick reference for the two platform waveforms. Open with GTKWave:

```
# Zynq-7000 (verified baseline)
gtkwave ppg_system.vcd hardware/zynq/presentation.gtkw

# ShrikeFi (ForgeFPGA + ESP32-S3)
gtkwave hardware/shrikefi/shrikefi_sim.vcd hardware/shrikefi/presentation.gtkw
```

Zynq-7000 — Key Signal Groups

Group	Signals	What to Watch
Clock/Reset	s_axi_aclk, s_axi_aresetn	50 MHz clock, active-low reset
AXI Write	awaddr, awvalid/awready, wdata, wvalid/wready, bvalid/bready	Register config: threshold (0x0C), etc.
AXI Read	araddr, arvalid/arready, rdata, rvalid/rready	Data extraction: filtered PPG (0x04/0x14), IBI (0x08), status (0x0C)
Filters	reg_red_raw, red_filtered, reg_ir_raw, ir_filtered	8-tap moving avg, 0 DSP, single-cycle latency
Peak Detector FSM	current_state (0=ARMED,1=RISING,2=PEAK,3=REFRACTORY), interval_cnt, refractory_cnt	4-state systolic detector, 250 ms refractory
Outputs	hw_beat_pulse, irq_beat, beat_flag, hw_ibi_cycles	Key result: IBI in 20 ns ticks
Scoreboard	tests_passed, tests_failed, test_num	Should show 6 passed, 0 failed

ShrikeFi — Key Signal Groups

Group	Signals	What to Watch
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Group	Signals	What to Watch
Clock/Reset	<code>clk</code> , <code>rst_n</code> (both TB and DUT)	ForgeFPGA clock domain
4-Bit Link	<code>link_strobe</code> , <code>link_dir</code> , <code>link_din[3:0]</code> , <code>link_dout[3:0]</code> , <code>link_dout_oe</code> , <code>strobe_sync</code>	Core innovation: 4-bit parallel FPGA↔MCU protocol
Filters	<code>reg_red_raw</code> , <code>red_filtered</code> , <code>red_valid_pulse</code> , <code>red_filtered_valid</code> (same for IR)	Same filter RTL, validated on ForgeFPGA
Peak Detector	<code>state</code> (link FSM), <code>u_peak_det.current_state</code> , <code>peak_beat_detected</code> , <code>reg_threshold</code> , <code>peak_ibi_cycles</code> , <code>reg_ibi_latched</code>	Peak detect + IBI latching for link transfer
Output	<code>irq_beat</code> (TB + DUT)	Beat interrupt to ESP32-S3
Scoreboard	<code>tests_passed</code> , <code>tests_failed</code> , <code>total_tests</code>	Should show 5 passed, 0 failed

Presentation Tips

1. **Zoom to first beat:** Both simulations show ~2-3 beats in ~2s sim time
2. **Color coding:** Groups are color-coded (Clock=blue, AXI=magenta/cyan, Filters=purple, FSM=red, Outputs=green, TB=orange)
3. **Cursor measurements:** Use GTKWave cursors to measure:
4. IBI cycles → heart rate: $HR = 60 / (ibi_cycles * 20ns)$
5. Filter latency: `data_valid` to `filtered_valid` = 1 cycle
6. **Export images:** File → Write Image → PNG for slides

Quick Signal Mapping (Zynq AXI ↔ ShrikeFi Link)

Zynq AXI Register	ShrikeFi Link Equivalent
<code>REG_RED_RAW</code> (0x00)	Nibble 0-1 of <code>link_dout</code> (FPGA→MCU)
<code>REG_RED_FILTERED</code> (0x04)	Nibble 2-3 of <code>link_dout</code>
<code>REG_IBI_CYCLES</code> (0x08)	Nibble 4-7 of <code>link_dout</code> (32-bit split)
<code>REG_STATUS_THRESH</code> (0x0C)	Status nibble in link protocol

Zynq AXI Register	ShrikeFi Link Equivalent
REG_IR_RAW (0x10)	Nibble 8-9 of link_dout
REG_IR_FILTERED (0x14)	Nibble 10-11 of link_dout

See docs/SHRIKEFI_LINK_PROTOCOL.md for full protocol spec.